

1 THE LANGUAGE-GROUNDED ROUTING PROBLEM

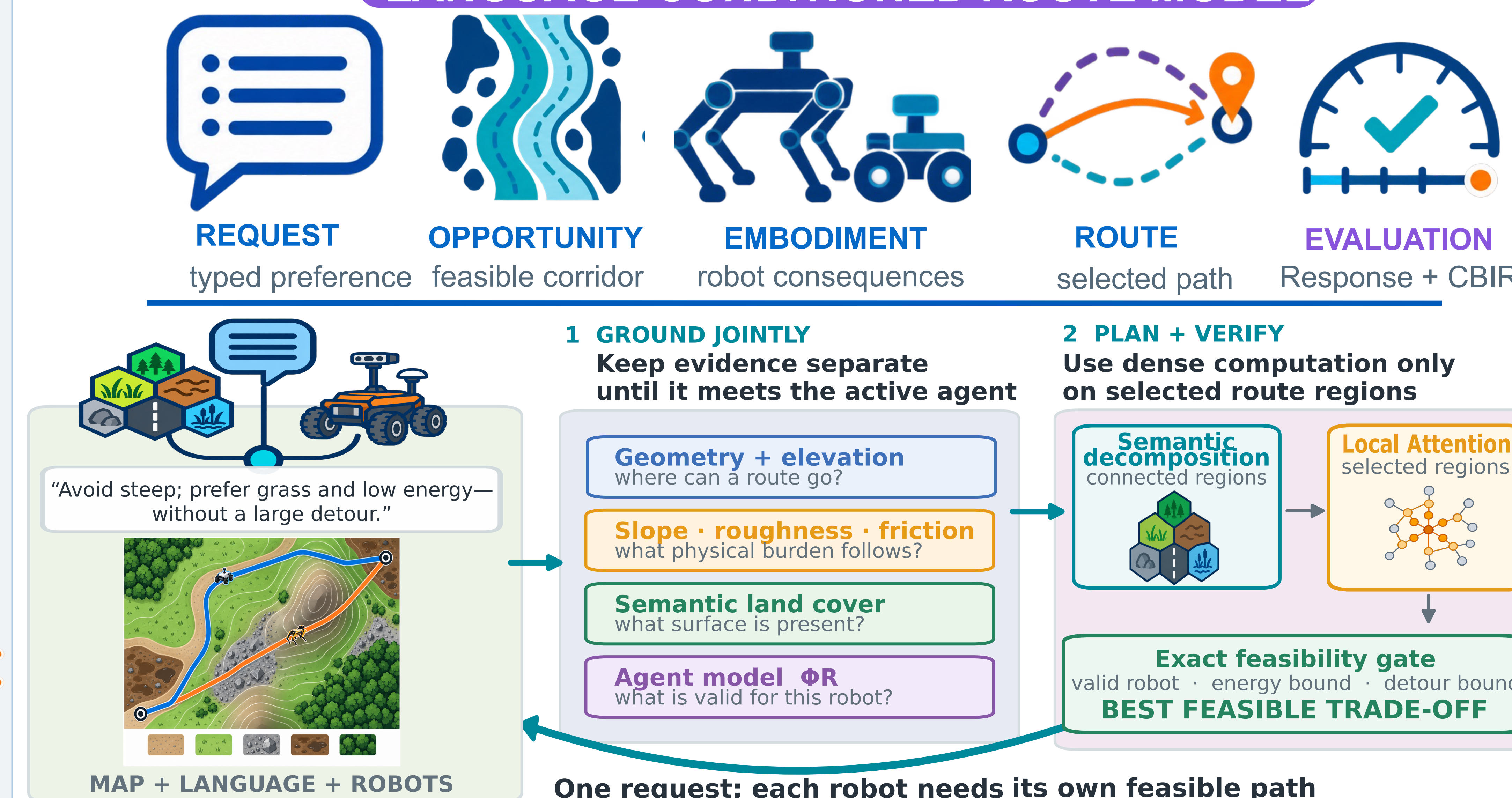
The same request can produce different safe routes.

Terrain opportunity changes across the map. Robot capability changes which alternatives remain feasible.

The planner must ground



LANGUAGE-CONDITIONED ROUTE MODEL

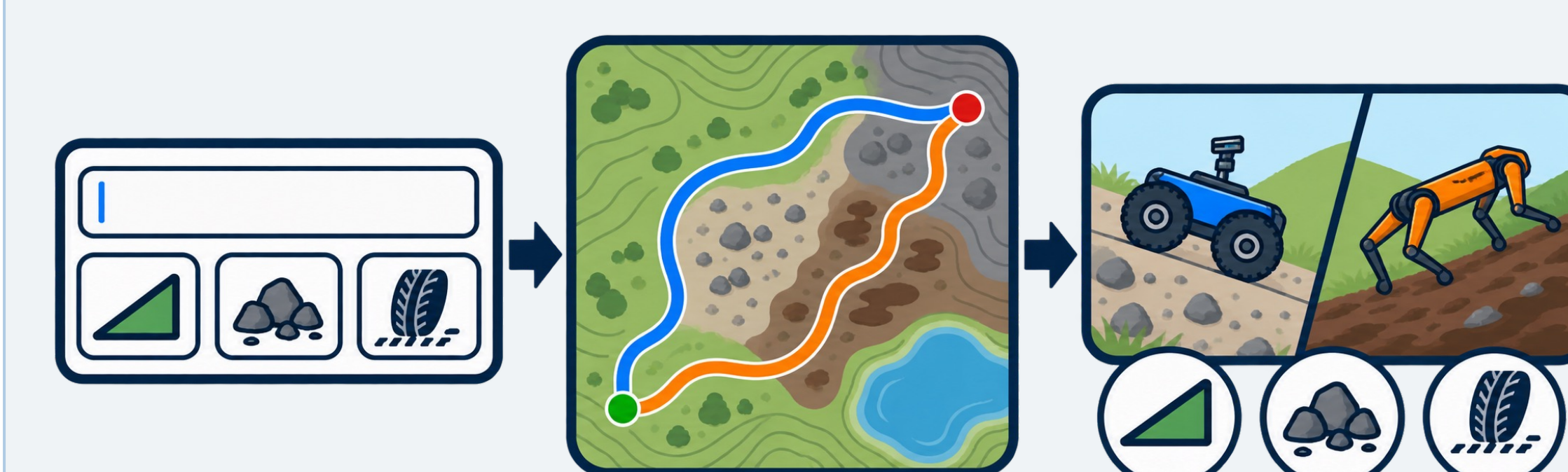


2 SCOPE

GLOBAL ROUTE SELECTION

Language mission preference

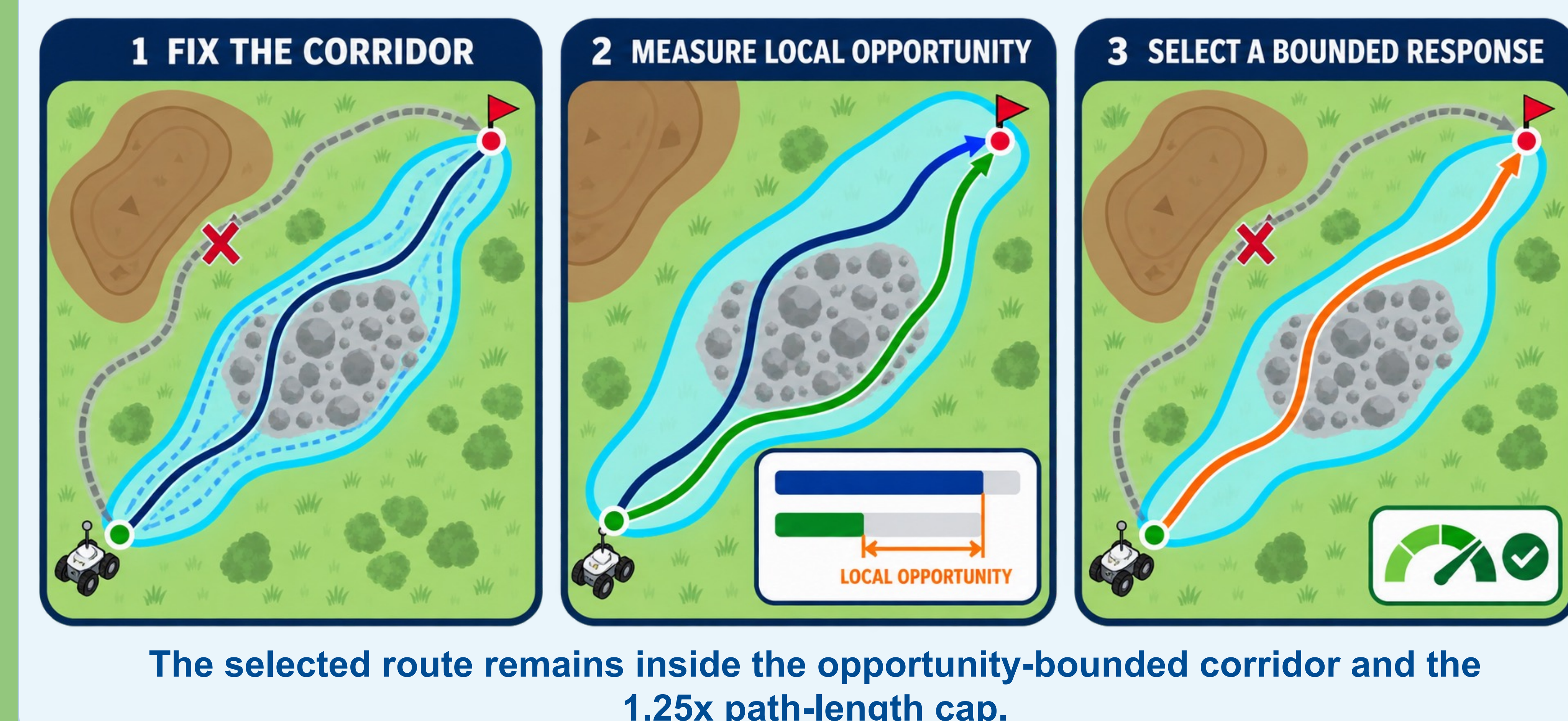
Held-out simulation evaluates slope, roughness, and traction intents for target platform.



3 FROM A TYPED REQUEST TO A CORRIDOR-FEASIBLE ROUTE

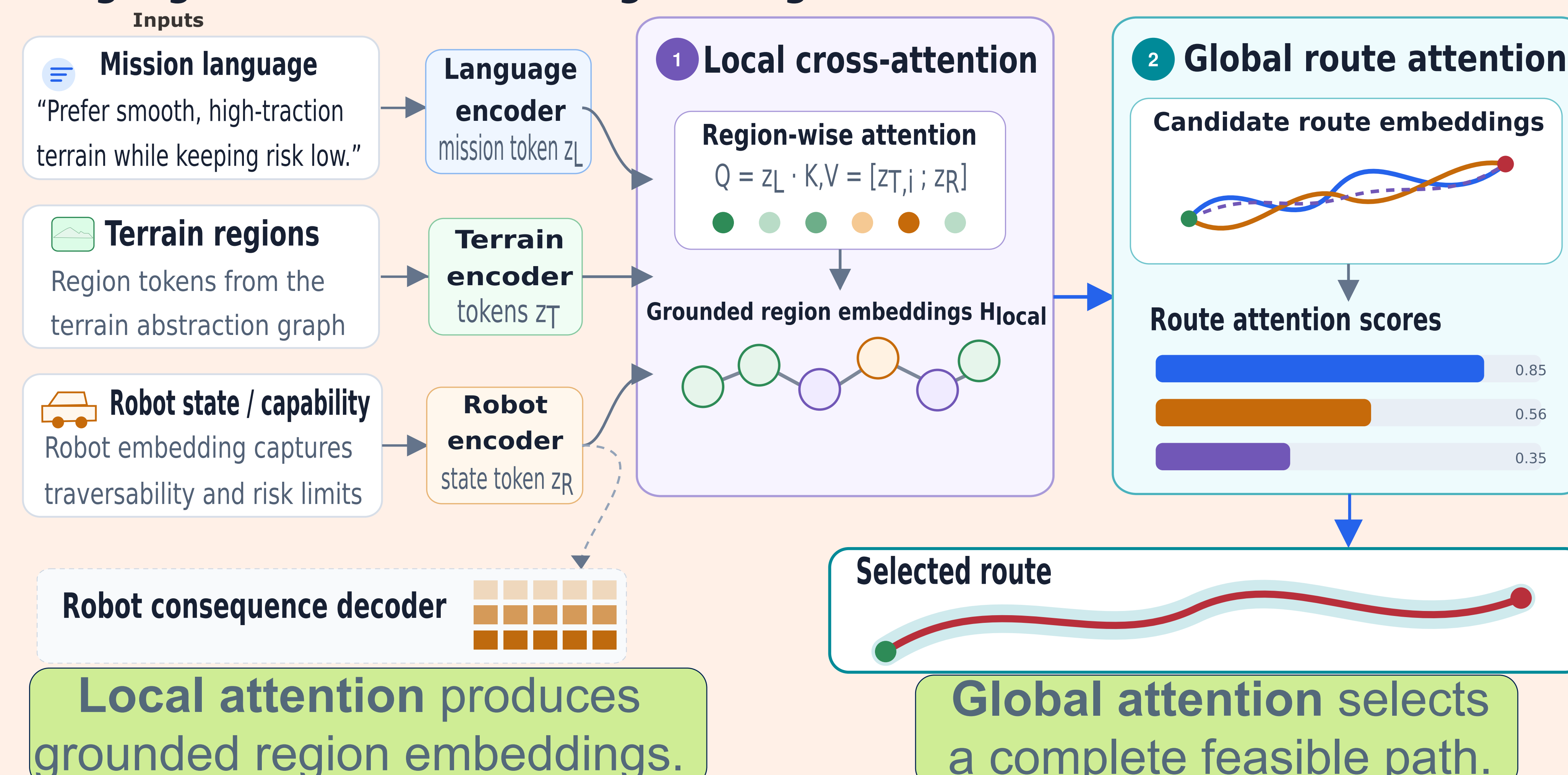
LOCAL OPPORTUNITY | BOUNDED CO`RRIDOR

Objective-specific lower bounds expose the feasible alternatives available for the request.



GLOBAL ROUTE SELECTION | SCORE COMPLETE CANDIDATES

Language-conditioned two-stage local-global attention



CORRIDOR CONTRACT GROUND requested terrain values EXPOSE feasible alternatives SCORE active robot consequences ABSTAIN when opportunity is insufficient

4 EVIDENCE ACROSS RESPONSE, SAMPLE SPACE, AND GROUNDED ROUTES

A EVALUATION METRICS

94.6%
RESPONSE

0.146
MEAN CBIR

95% CI 93.4–95.8%
n = 2,519 actionable atomic cases

95% CI 0.133–0.160
n = 2,505 referenced cases

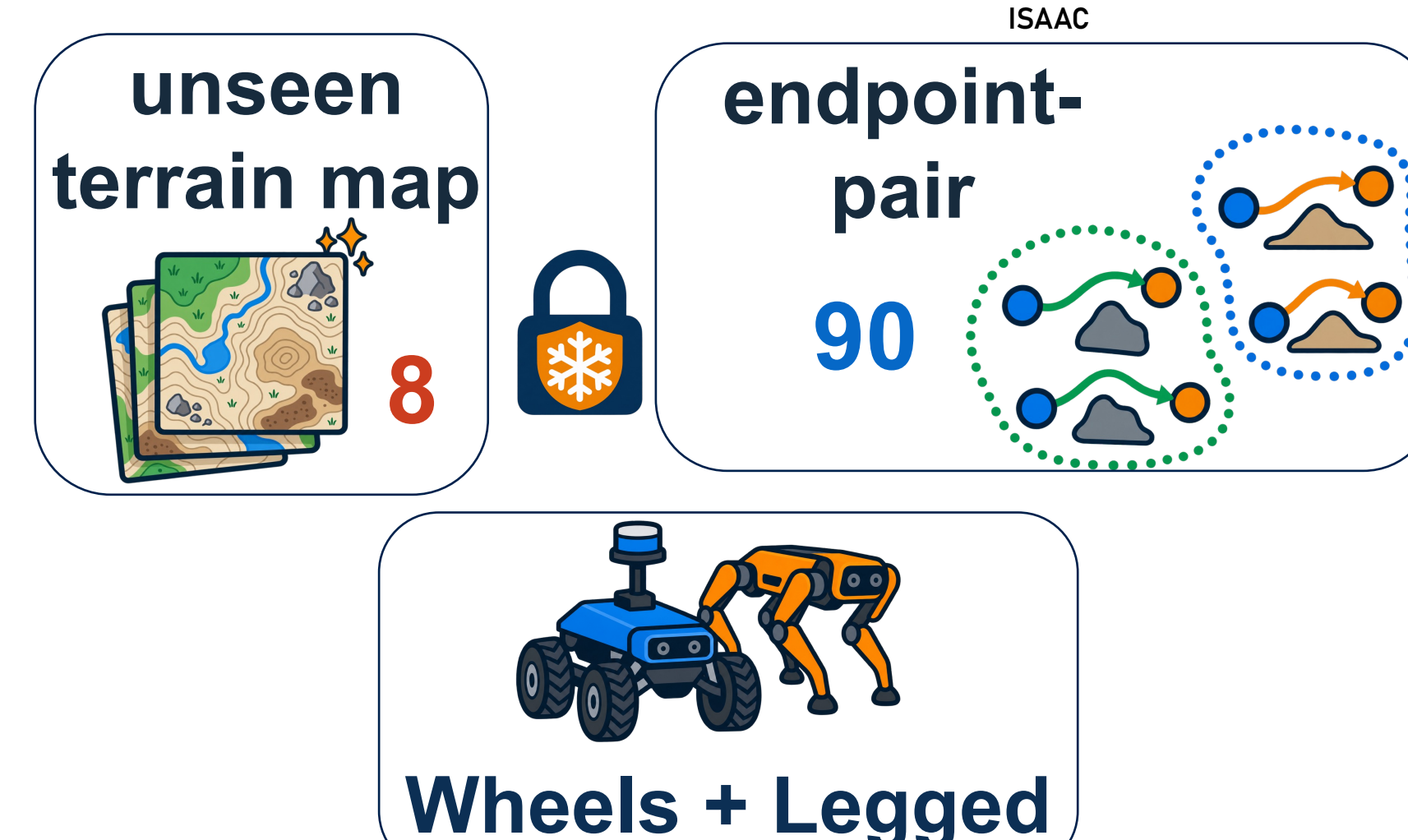
$Response_i = \mathbb{I}[gain_i \geq \max(0.005, 0.10 opportunity_i)]$

Response tests whether movement passes the opportunity-scaled threshold.

$$CBIR_j(\hat{p}) = \frac{g_j(\hat{p}) - g_j(p_{ref})}{g_j(p_0) - g_j(p_{ref})}$$

CBIR: 0 = reference, 1 = neutral
lower is better.

B Dataset

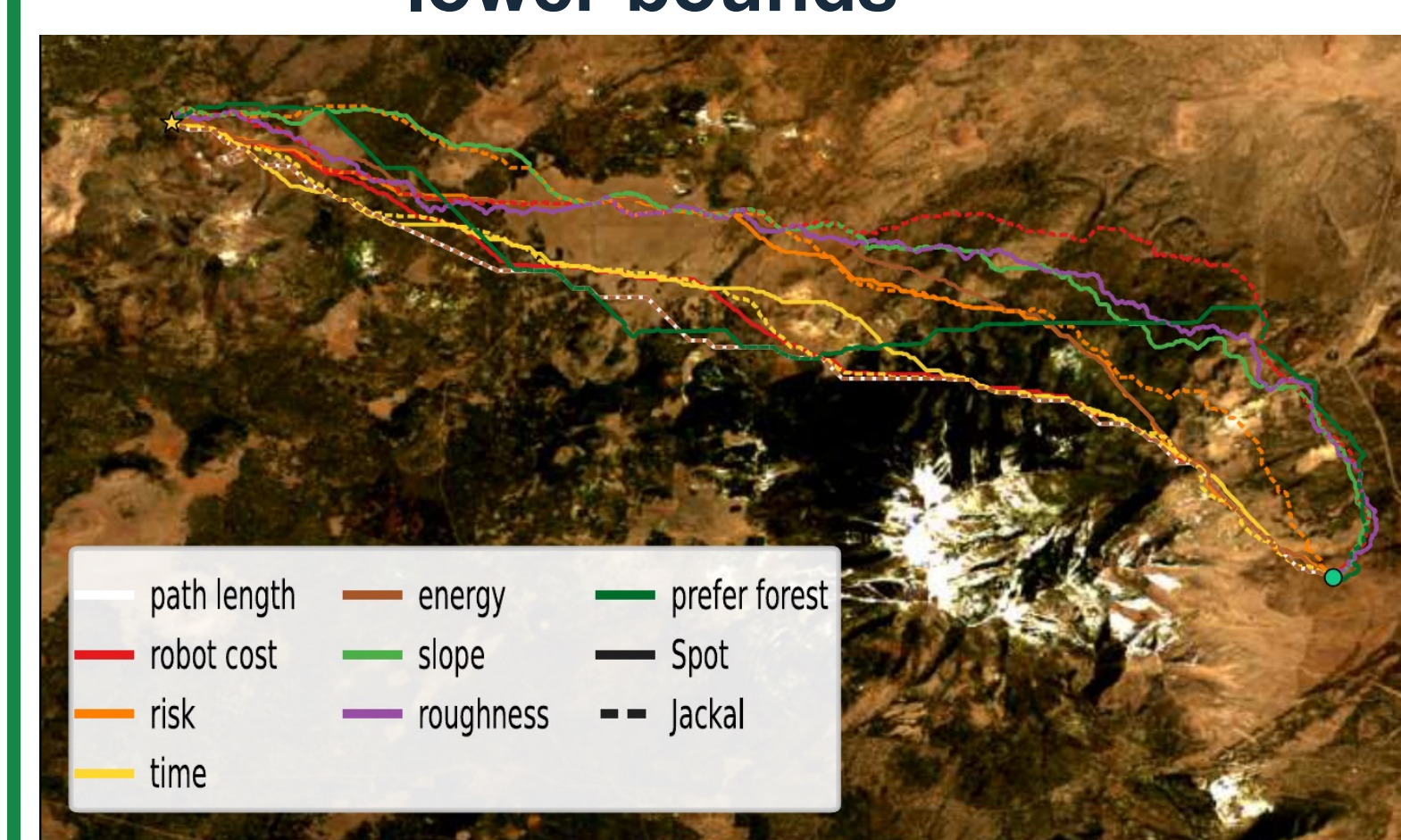


All selected routes satisfied the simulated energy budget.

C VISUAL RESPONSE CASES

Lower Bound
Corridor Paths

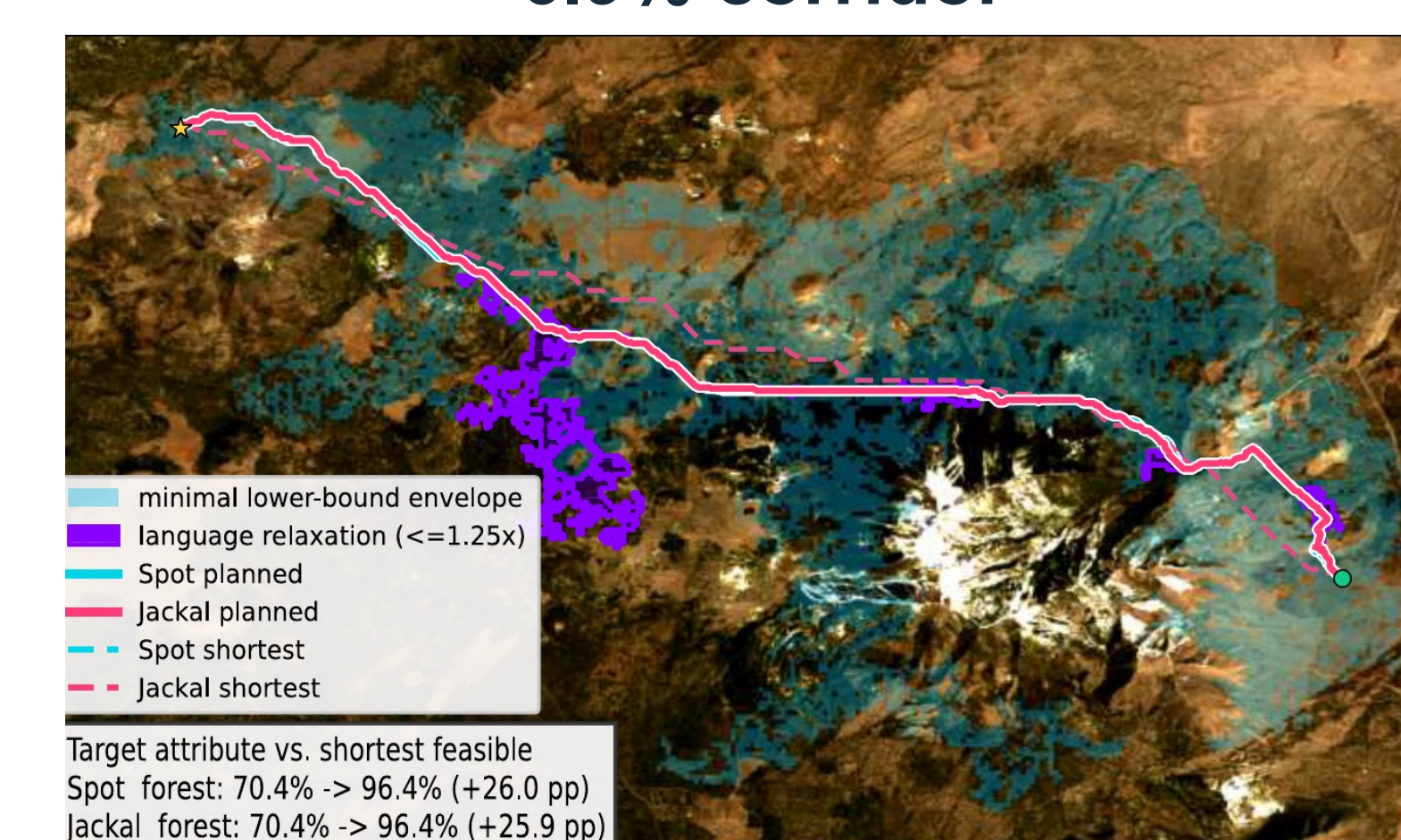
Robot-specific feasible lower bounds



TERRAIN INTENT

"Prefer forested terrain while remaining robot-feasible."

Planned Route within 6.9% corridor

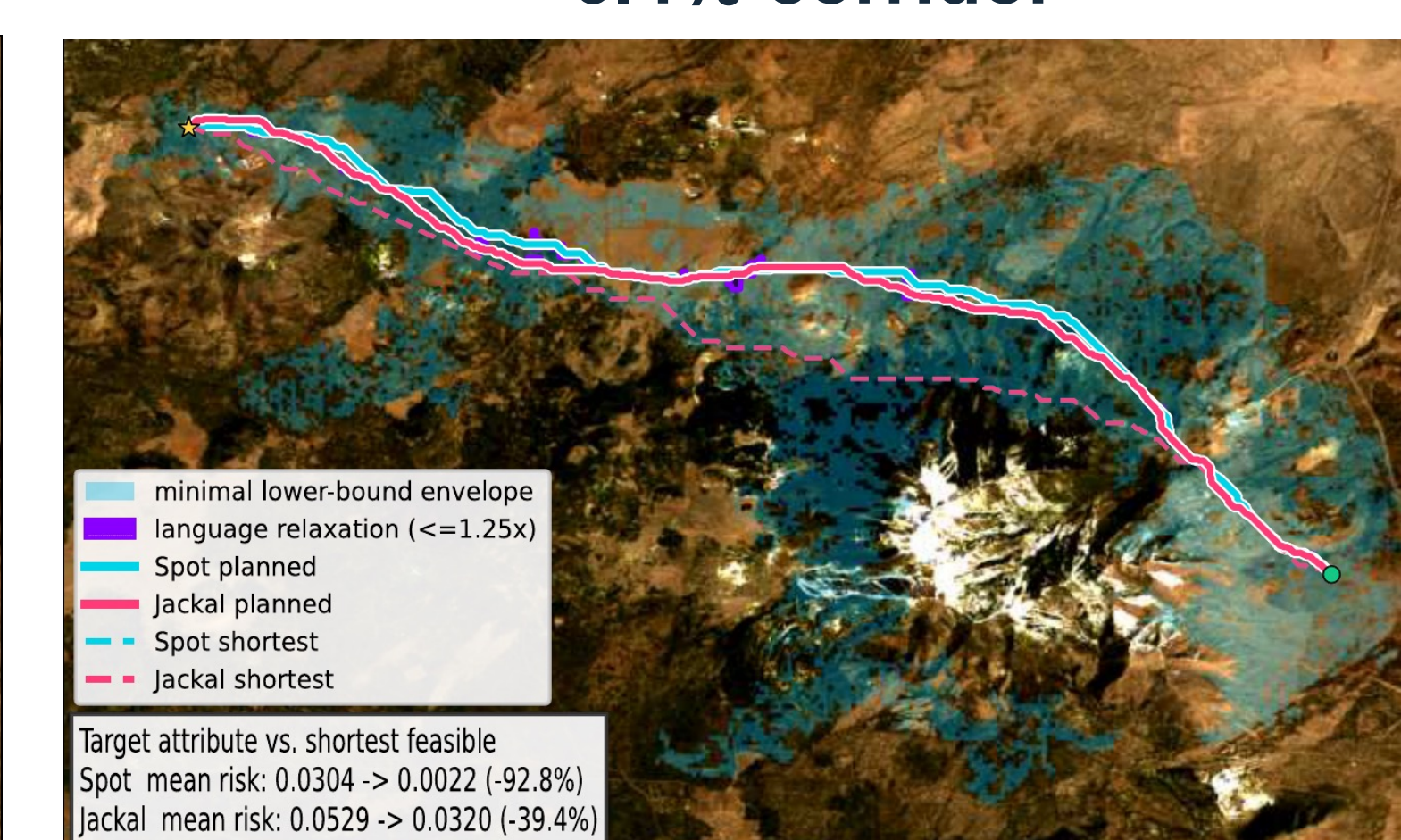


Both robots increase forest exposure while remaining near feasible length lower bounds.

ROBOT-CONDITIONED RISK

"Minimize traversal risk for the active robot"

Planned Route within 6.4% corridor



Spot and Jackal select different routes inside their own feasible corridors.

Metrics begin from structured intents and do not measure free-form language accuracy.